Team Details

- ➔ Team name – TeamMAGNUS
- ➔ Team leader name – Sumit Mondal
- ➔ Problem Statement - Travelers often use multiple disconnected apps for navigation, weather updates, traffic monitoring, accommodation booking, restaurant search, fuel stations, mechanic support, and sightseeing discovery. Existing platforms do not integrate all travel needs into a single intelligent ecosystem, leading to inefficient planning, safety risks, and missed exploration opportunities along travel routes.

Brief about the idea

- Our solution is an AI-powered smart travel ecosystem designed for all types of travelers — including bike riders, car travelers, road-trippers, campus groups, and solo explorers.
- The platform integrates real-time weather updates, traffic analysis, fuel prediction, nearby stay recommendations, restaurant suggestions based on time of day (breakfast, lunch, evening snacks, dinner), nearby mechanic and fuel stations, safety-based route risk scoring, and dynamic sightseeing discovery along travel routes — all within a single intelligent application.
- Unlike traditional navigation apps, our system acts as a complete travel assistant by combining AI-driven optimization with community-powered insights. It ensures safer journeys, smarter budgeting, and enriched exploration of attractions located along the road.
- The platform uses AI-driven personalization to learn user preferences such as travel style, budget range, food preferences, safety sensitivity, and sightseeing interests. Based on past trips and behavior patterns, it dynamically recommends optimized routes, suitable restaurants, ideal stay options, and customized sightseeing stops — making every journey uniquely tailored to the traveler.

Opportunities:-

- ➔ Unlike existing travel apps that focus only on navigation, booking, or food discovery separately, our platform integrates real-time weather, traffic analysis, nearby stays, time-based restaurant suggestions, mechanic support, fuel alerts, and route-based sightseeing discovery into a single AI-powered ecosystem. It provides a complete travel assistant rather than a single-feature application.

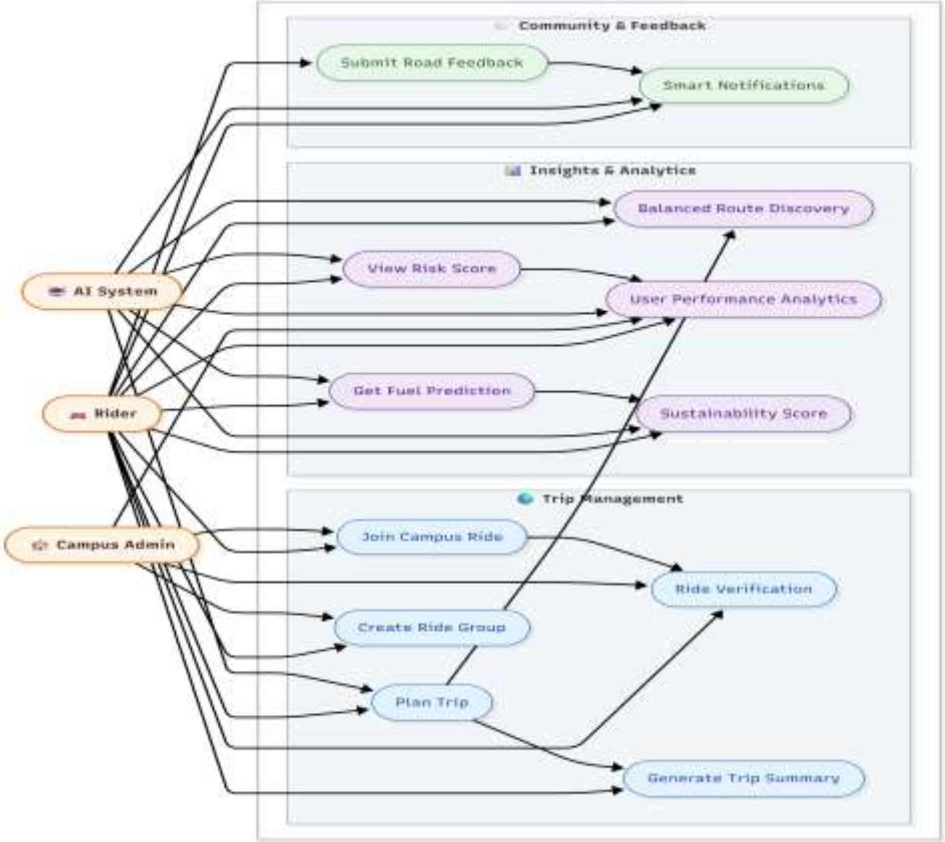
- ➔ The solution reduces travel complexity by combining multiple travel services into one intelligent platform. It optimizes routes, provides safety alerts, suggests nearby essential services, and highlights sightseeing spots along the journey, ensuring safer, smarter, and more enjoyable travel planning without switching between multiple apps.

- ➔ An all-in-one AI-powered travel intelligence ecosystem that integrates navigation, safety monitoring, service discovery, personalization, and route-based exploration into a unified platform for all types of travelers.

List of features offered by the solution:-

- Smart Multi-Mode Route Planner (Bike, Car, Group Travel)
- AI-Based Route Optimization (Time, Cost, Safety Balanced)
- Real-Time Weather Updates & Alerts
- Traffic Congestion Analyzer
- Fuel Prediction & Nearby Fuel Station Locator
- Nearby Stay Recommendations (Hotels, Hostels, Camping Options)
- Time-Based Restaurant Suggestions (Breakfast, Lunch, Evening Snacks, Dinner)
- Nearby Mechanic & Emergency Support Locator
- Route Risk Scoring System
- Dynamic Sightseeing Discovery Along the Travel Route
- Personalized Travel Recommendation Engine
- Community Reviews & Safety Feedback System
- Group & Campus Trip Coordination Tool
- Carbon Footprint Tracker
- Offline Mode for Remote or Low-Network Areas
- Auto Trip Summary & Shareable Report Generator

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

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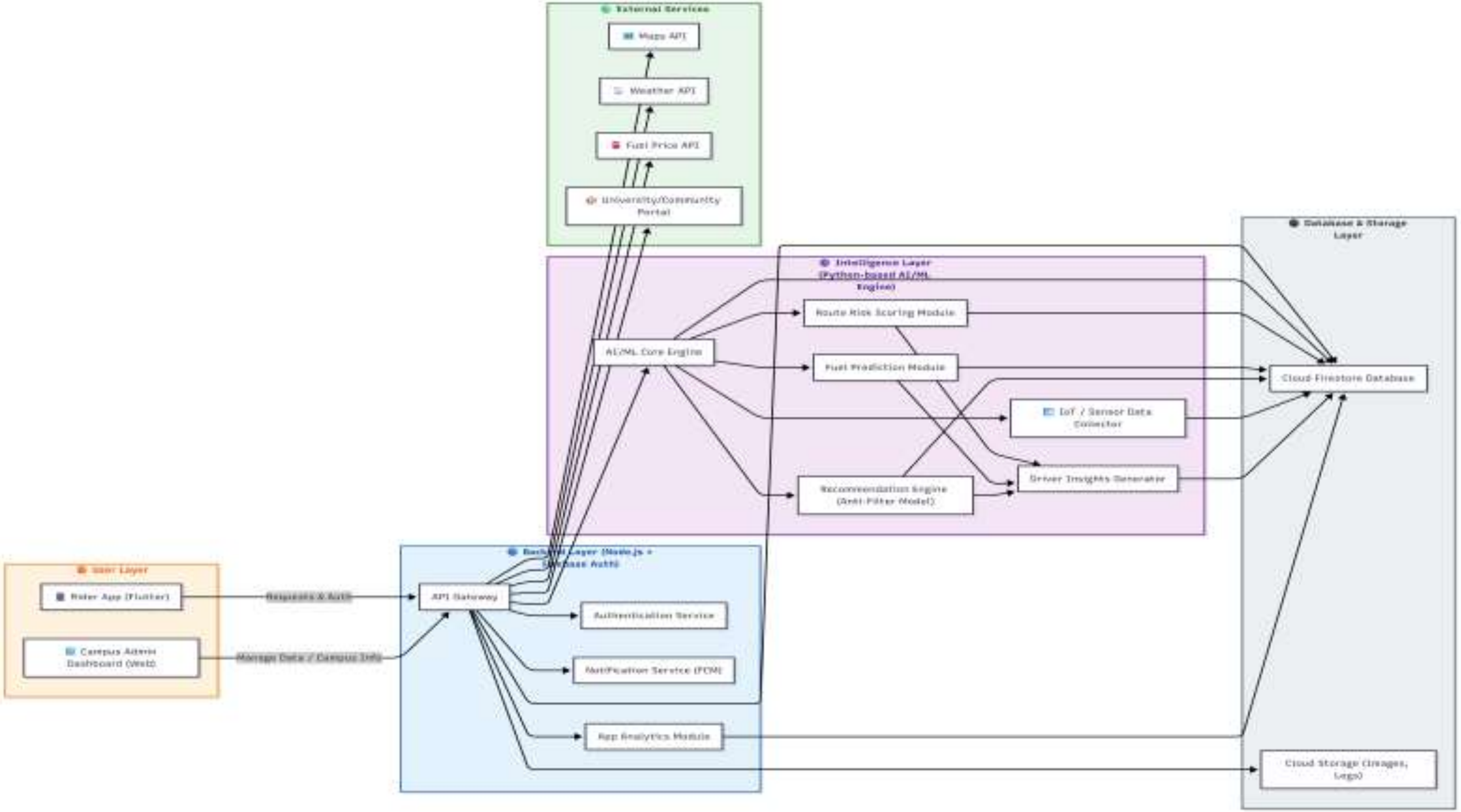
Architecture diagram of the proposed solution



Technologies to be used in the solution

- **Firebase** – Authentication, Cloud Firestore Database, and Real-time Data Sync
- **Node.js** – Backend server and API management
- **Python** – AI/ML models for route optimization, risk scoring, and personalization
- **Machine Learning Algorithms** – Recommendation systems and predictive analytics
- **RESTful APIs** – Integration with external services
- **Maps API (Google Maps / OpenStreetMap)** – Navigation and geolocation services
- **Weather API Integration** – Real-time weather updates and forecasts
- **Places API** – Nearby restaurants, stays, sightseeing spots, mechanic shops
- **Geospatial Data Processing** – Route analysis and point-of-interest detection
- **Cloud Hosting (Firebase / Cloud Platform)** – Scalable deployment
- **Push Notification Service** – Real-time alerts and emergency notifications
- **Data Analytics Tools** – Travel insights and performance monitoring
- **Offline Data Caching Technology** – Support in low-network areas

Usage of AMD Products/Solutions

- **AMD Ryzen™ Processors** – Used for high-performance development, backend processing, and training AI/ML models for route optimization, traffic analysis, and risk scoring.
- **AMD Radeon™ GPUs** – Utilized for accelerating machine learning computations, geospatial data processing, and real-time route analytics.
- **AMD-Powered Systems for Model Training** – Heavy AI workloads such as predictive weather modeling, personalization engines, and traffic data analysis can be optimized using AMD's multi-core architecture for faster computation.
- **Parallel Processing with AMD Architecture** – Enables efficient handling of large-scale travel datasets, including route mapping, location intelligence, and real-time recommendation systems.
- **Scalable Deployment & Testing** – Application backend and simulation testing can be optimized on AMD-powered cloud or workstation environments for performance efficiency.

Estimated implementation cost (optional)

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Prototype Assets (Optional)

➔ GitHub Public Repository Link:-https://github.com/Sumit648527/AMD

➔ Demo Video Link:- https://drive.google.com/drive/folders/1jiCzmENKc1luaTdsBxFibxbZekCuKcde?usp=sharing

Add as per the requirements of the contest

Our proposed solution aligns with the contest objectives by delivering an AI-driven, scalable, and high-performance travel intelligence platform that integrates real-time mobility analytics, predictive modeling, and personalized recommendation systems. The application leverages advanced computing capabilities for route optimization, weather-risk analysis, traffic modeling, and geospatial data processing.

The system is designed to:

- Demonstrate innovation in smart mobility and intelligent travel planning
- Utilize AI/ML models accelerated on AMD-powered systems
- Showcase scalable cloud-based architecture
- Address real-world travel planning challenges with practical implementation
- Promote sustainable mobility through carbon tracking and optimized routing

The prototype will include a working mobile interface, integrated APIs, and an AI-based route intelligence engine to demonstrate feasibility and technical depth, ensuring alignment with the technical and innovation benchmarks of the competition.

Our project is an AI-powered Smart Travel Intelligence Ecosystem designed for all types of travelers. Many travelers currently depend on multiple disconnected applications for navigation, weather updates, traffic monitoring, accommodation booking, food discovery, fuel stations, and sightseeing information, which makes planning inefficient and complicated. Our solution integrates all these features into one unified platform. It offers AI-driven route optimization, real-time weather and traffic insights, time-based restaurant suggestions, nearby stay recommendations, mechanic and fuel station locators, and dynamic sightseeing discovery along travel routes. Built on scalable architecture and high-performance computing, the system enhances safety, personalization, convenience, and overall travel efficiency, creating a smarter and more seamless travel experience.

AMD

Slingshot

HUMAN *IMAGINATION*

BUILT WITH *AI*

Powered by H2S

Thank you!